train_biases.py

`scripts/train_biases.py · 187 lines`

What it does

Fine-tunes only the bias vectors (bias method) so each pruned layer reproduces the original layer's output on the calibration training split, monitoring the validation split.

Run

```
uv run python scripts/train_biases.py
```

Pipeline order: inspect_model -> build_c4_support_eval -> collect_stats -> score_experts -> replace_experts -> evaluate (before_training) -> train_biases -> evaluate (after_training) -> export_model

Reads

- `replaced-model/`
- calibration data
- the original model (once, to cache layer references)

Writes

- `<run_dir>/trained-model/`
- `training_report.json`
- layer reference cache in `CACHE_DIR`

Notes

- Verifies that only bias vectors are trainable and that routers and kept experts are unchanged afterwards.
- Exits immediately for `drop` and `reroute`, which have nothing to train.

Defined in this file

Extracted from the source code (classes, public functions and methods).

Name	Description
`main()`	Script entry point: runs this stage end to end.

Settings it reads

`BIAS_BATCH_SIZE`, `BIAS_EVAL_INTERVAL`, `BIAS_LEARNING_RATE`, `BIAS_TRAINING_OBJECTIVE`, `BIAS_TRAINING_STEPS`, `CALIBRATION_BATCH_SIZE`, `CHECKPOINT_FORMAT`, `MODEL_DTYPE`, `MODEL_PATH or HF_MODEL_ID`, `REPLACEMENT_METHOD`, `SEED`

Connections

Uses: `moe_pruning.data.calibration`, `moe_pruning.model.architecture`, `moe_pruning.model.loader`, `moe_pruning.pruning.bias_expert`, `moe_pruning.pruning.replacement`, `moe_pruning.serialization.checkpoint`, `moe_pruning.training.bias_trainer`, `moe_pruning.utils.cli`, `moe_pruning.utils.device`, `moe_pruning.utils.paths`, `moe_pruning.utils.reproducibility`